MCO-1D

Test Cards

Structured test procedures for commissioning and first-flight envelope

v25 · September 2026

v25 additions: TC-REGEN-1 (regenerative energy recovery bench test), TC-ENVELOPE-1 (first-flight envelope characterisation procedure), TC-WEAR-1 (wear baseline validation after 3 flights).

TC-REGEN-1 Regenerative energy recovery bench test

Verify ODrive dc_bus_max_negative_current = –5.0 on all 8 axes. With aircraft restrained, drive wings at cruise speed then command active brake zone. GS: regen_status should show regen_active=True and I_regen_A > 0. Verify no overvoltage faults on ODrive. Pass criterion: regen_active confirmed, no faults, bus voltage stable.

ODrive config	dc_bus_max_negative_current = –5.0 on all 8 axes
Test condition	Restrained, cruise speed, then lever to 10%
Pass	regen_active=True, I_regen_A > 0, no ODrive faults
Fail action	Check wiring, re-run regen_config_advice
Safety	Battery <95% SOC during test. Remove regen config if faults

TC-ENVELOPE-1 First-flight envelope characterisation

H-PERF-6 runs automatically. Pilot executes the deceleration survey and hover attempt per Commissioning Manual section 3. Pass criterion: HOVER_CAPABLE and V_MIN_MS populated in _ENVELOPE. All subsequent flights conditioned on these values.

Pre-condition	First flight only. H-PERF-6 at queue position 0.
Pilot action	Decelerate in 0.5 m/s steps, hold 10s each. Attempt hover.
Pass	_ENVELOPE populated: HOVER_CAPABLE = True or False, V_MIN_MS set
Post	set_hover_capable, alc_debrief, persist_save_all
Note	If aircraft cannot slow below 4 m/s: forward-flight SAR mode active

TC-WEAR-1 Wear baseline validation — post flight 3

After the third flight, check wear_status shows baseline_set = True and per-wing baseline_init_I, baseline_mid_I values are populated. Compare port vs starboard baseline symmetry. Asymmetry > 20% may indicate servo calibration issue.

Pre-condition	At least 3 flights completed, persist_save_all after each
Check	wear_status: baseline_set = True, all 8 wings have baseline values
Symmetry check	baseline_init_I values within 20% port vs starboard
Pass	Baseline set, all 8 wings, port/stbd symmetric
Action on fail	Re-check servo calibration, re-run wear collection flight

TC-FF-SAR-1 Forward-flight SAR mode validation

If HOVER_CAPABLE = False: validate that H-SAR-FF-1 through H-SAR-FF-4 are active and that hover-dependent hypotheses are gated. Fly a slow orbit at V_min and confirm sensor pointing geometry.

Pre-condition	HOVER_CAPABLE = False set from TC-ENVELOPE-1
Check GS	envelope_status: hover-dependent hyps gated
Fly test	Slow orbit at V_min, 15° bank, 30m AGL
Measure	Orbit radius (should match $V^2/(g \times \tan(15°))$)
Pass	Aircraft maintains orbit, sensor points toward target area
Record	GS: envelope_status → orbit_radius_m for mission planning